

Dense Drawdown Supervision for Equity Crisis Early Warning: A Multi-Task Temporal Fusion Transformer and an Honest Test of Regime Awareness

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Abstract

Single-market equity crisis early-warning systems (EWS) are data-starved and gradient-boosted trees usually win. Taking the Temporal Fusion Transformer (TFT) as a backbone, we test, under a leak-free walk-forward protocol with a 63-day embargo and paired significance tests, whether *regime awareness* (a differentiable HMM regime module + regime-conditioned attention bias) helps. It does not: a fully regime-aware TFT is indistinguishable from a vanilla TFT ($p=0.43$). What helps is a multi-task *auxiliary forward-drawdown head* + *focal loss* (“AF”): it beats vanilla TFT single-market (macro PR-AUC 0.483 vs. 0.458, $p=0.027$) with far better calibration, and, scaled to a nine-market panel (≈ 110 episodes), beats XGBoost (PR-AUC 0.399 vs. 0.373). Dense supervision beats architectural ornament; protocol beats headline.

1 Problem and approach

A reliable early-warning system (EWS) for equity crises is of direct value to risk managers and allocators (Kaminsky & Reinhart, 1999). We predict crisis *onset* within a forward quarter $[t+1, t+63]$. The literature reports that on single markets gradient-boosted trees match or beat neural sequence models (Bluwstein et al., 2023; Holopainen & Sarlin, 2017); this is a property of the *formulation* (few episodes), not a law. We take the TFT (Lim et al., 2021) as backbone (Fig. 2) and ask two questions: does a crisis-specific inductive bias — *regime awareness* — help, and if not, what does?

Contributions. (i) A leak-free evaluation that makes an earlier random-holdout “win” reverse. (ii) A powered, honest test showing the regime machinery is *inert*. (iii) The AF recipe: dense drawdown supervision + focal loss, which significantly and transferably improves on vanilla TFT.

2 Data and labelling

Daily Bloomberg data, 2000–2026: the S&P 500 (single-market study) and a nine-index panel (SPX, NDX, RTY, SX5E, SXXP, UKX, MCX, MXEF, MXWO; 61,996 market-days, ≈ 110 crisis episodes vs. ~ 5 on the S&P 500 alone). Inputs are a 252-day window of 23 historical features (multi-horizon returns, realised vol, drawdown, Hurst, Amihud illiquidity, VIX, credit/TED spreads, DXY, gold, MOVE, ...) plus two known-future calendar features. The dataset has *no* static numerics, so TFT’s static pathway activates only in the panel, via a learned market embedding.

Crisis = drawdown $|D_t| \geq \tau$ from the trailing 252-day peak; onset = first crisis day after ≥ 20 calm days; target $y_t = \mathbf{1}\{\text{onset in } [t+1, t+63]\}$. We choose $\tau=10\%$ from the empirical drawdown distribution (mean 6.6%, s.d. 9.0%): 5% is below the mean (-0.18σ , noise), 20%

is clean ($+1.49\sigma$) but yields ~ 5 episodes; dd10 ($+0.38\sigma$, 23.6% of days) is the defensible operating point for a market-stress EWS.

3 Method

Leak-free protocol. Crisis-anchored walk-forward folds (Fig. 1); a 63-day embargo purges the label-horizon rows at each boundary (López de Prado, 2018); train/val/test are windowed independently (strictly post-split); the scaler is train-only. Primary metric PR-AUC (read against prevalence (Saito & Rehmsmeier, 2015)); also ROC-AUC, Brier, Platt calibration; 30 seeds; paired Wilcoxon tests.

The AF recipe. Two changes to the *learning problem*, not the topology: a multi-task linear head regressing the continuous forward drawdown (a dense signal present every day, unlike the sparse onset label), and focal loss (Lin et al., 2017) on the classifier. With predicted probability p , label y , predicted/target drawdown $\hat{d}, d$:

$$\mathcal{L} = -w_+ y(1-p)^\gamma \log p - (1-y)p^\gamma \log(1-p) + \lambda(\hat{d} - d)^2, \quad (1)$$

with $\gamma=2$, $\lambda=0.5$, w_+ near the inverse prevalence. “AFv2” is AF with panel-matched capacity (hidden 128, 60 epochs, lr 5×10^{-4} , per-market norm).

4 Results

Backbone. On clean splits, vanilla TFT (0.515) > LSTM (0.397) > XGBoost (0.320) macro PR-AUC — the earlier “LSTM wins” was a leakage artifact.

Regime awareness is inert. Factorially ablated, no component helps; the full RA-TFT ties vanilla (Table 1; $p=0.43$, 30 seeds, 6 folds). Adding seeds cannot fix this — power is bounded by the number of independent crisis *episodes*, not seeds.

The AF recipe wins. AF beats vanilla TFT on every metric and is the *first* configuration to do so significantly on clean splits (Table 1, $p=0.027$); its calibration edge (Brier 0.270 vs. 0.318) is the clearest practical gain. The improvement concentrates in severe regime-shift crises (COVID +0.16, China +0.04 PR-AUC) and is neutral on mild corrections — it helps most when it matters most.

Panel. With capacity matched to the larger data, AFv2 beats XGBoost (Table 2): PR-AUC 0.399 vs. 0.373, ROC-AUC 0.609 vs. 0.585. The regime module is inert again ($p=0.95$, a fifth confirmation); the lever is capacity+epochs+loss, not per-market norm ($p=0.46$). The same AFv2 config transfers back to the single market without loss — one unified recipe.

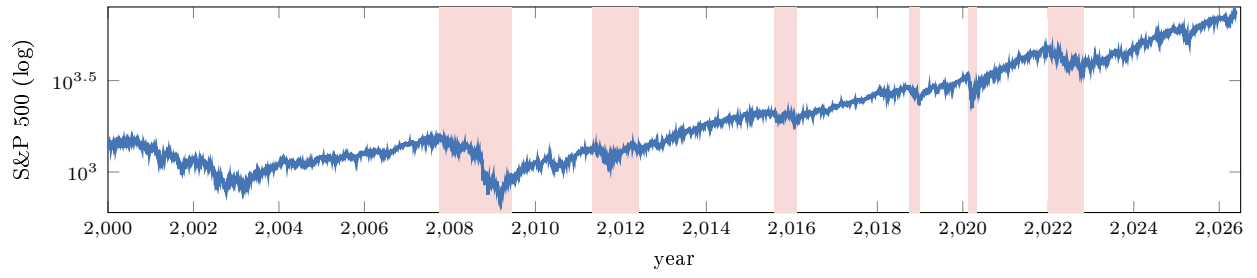


Figure 1: S&P 500 (log), 2000–2026, with the six crisis episodes used as walk-forward test folds shaded: GFC, Eurozone, China-2015, Q4-2018, COVID, 2022 bear. Each fold trains on the expanding prior history (with a 63-day embargo) and tests on the shaded crisis.

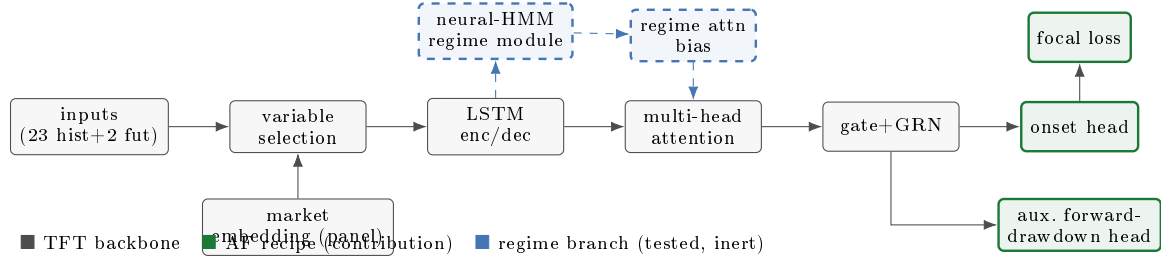


Figure 2: Model. Black: TFT backbone (Lim et al., 2021). Green: the AF recipe (multi-task auxiliary forward-drawdown head + focal loss), the contribution. Blue/dashed: the regime module and regime-conditioned attention bias, implemented and ablated but found inert.

Table 1: Single-market, 6 folds, ≈ 30 seeds. Macro PR-AUC (raw/Platt), Brier $\downarrow$. AF vs. vanilla $p=0.027$; RA-TFT vs. vanilla $p=0.43$.

Model	raw PR-AUC	Platt PR-AUC	Brier	Platt Brier
Vanilla TFT (M0)	0.458	0.545	0.318	0.285
Full RA-TFT (M4)	0.460	0.623	0.305	0.270
AF (aux+focal)	0.483	0.656	0.270	0.239

Learning. Wiley.
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Table 2: Nine-market panel, pooled test (2019–2026), prevalence 0.300. AFv2 vs. vanilla+norm $p=0.039$.

Model	raw PR-AUC	ROC-AUC	Brier
AFv2 (tuned recipe)	0.399	0.609	0.269
M0 + per-market norm	0.377	0.597	0.347
XGBoost	0.373	0.585	0.273
Vanilla TFT (M0)	0.343	0.552	0.368
LSTM	0.297	0.490	0.478

5 Conclusion and future work

A well-motivated crisis-specific inductive bias (regime awareness) does not survive a powered, leak-free test, while a simple change to the learning problem — dense drawdown supervision + focal loss — does, transferably, across single-market and panel settings, beating XGBoost on the panel. Future work: more crisis episodes (the binding constraint) via a wider panel; instrumented evaluation for reliability diagrams and block-bootstrap CIs; and auxiliary-task design (multi-horizon / quantile drawdown heads).

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